Written Assignment 01

Due Wednesday, March 18, 2026 at 11:59 PM

Name: _____
Course / Section: _____
Chosen Language: _____

In this exercise, you will study and analyze a compiled programming language of your own choosing. The purpose of this activity is to help you understand why the language was created, how it works internally, and what kinds of problems it is designed to solve.

Rather than focusing on syntax alone, this exercise emphasizes language design, technical characteristics, and practical relevance.

INSTRUCTIONS

1. Answer all questions with respect to your chosen language.
2. All answers must be **handwritten** and written in complete sentences.
3. Do **not** copy definitions directly from websites or textbooks.
   Write in your own words and explain concepts clearly.
4. Print this document. Use additional sheets if necessary, but write your answers under the corresponding question number.

Historical Context and Motivation

1. Identify the programming language you selected. State its creator(s) and the year of its initial release.

2. Describe the historical context in which the language was developed. What limitations or deficiencies of existing languages motivated its creation?

3. Explain the primary goals of the language's designers. What problem domain or system requirements was the language intended to address?

Design Philosophy and Principles

4. Describe the core design philosophy of the language. What guiding principles influenced its design?

5. Explain how this philosophy is reflected in:

   (a) the language's syntax

   (b) its type system

   (c) its compilation strategy

6. Identify and discuss at least one major design trade-off made by the language designers. Explain what was gained and what was sacrificed.

Technical Characteristics

7. Describe the compilation and execution model of the language. Trace the process from source code to executing machine code.

8. Characterize the language's type system. Is it static or dynamic? Strong or weak? How does this affect program correctness and safety?

9. Explain the memory management strategy used by the language.

10. Describe the language's support for concurrency or parallelism. If such support exists, explain the mechanisms provided. If not, explain how concurrency is typically handled.

Comparative Analysis

11. Compare your chosen language with one other compiled language. Identify:

    (a) one significant similarity

    (b) one significant difference

12. Based on your comparison, explain why a programmer might prefer your chosen language over the alternative.

Evaluation and Practical Relevance

13. Describe the most common contemporary use cases of the language.

14. Identify the types of systems or problem domains for which the language is particularly well suited.

15. Discuss the principal strengths of the language.

16. Discuss the principal weaknesses or limitations of the language.

17. Evaluate the current relevance of the language. Is it still appropriate for modern software development? Justify your answer.

Reflection and Synthesis

18. Identify one concept related to this language that you understand more clearly as a result of this exercise. Explain it in your own words.

19. Suppose you were to design a new programming language. Which idea from your chosen language would you adopt, and which would you modify or reject? Explain your reasoning.

20. Provide a concluding paragraph (5–7 sentences) summarizing why the language was created and what types of problems it is fundamentally intended to solve.